

Advancing Urdu Toxicity Detection: Improved Span Models and a Multimodal Audio Extension

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Abstract

The proliferation of toxic content on Urdu social media platforms presents a significant challenge for automated content moderation, particularly given the morphological complexity, dual-script usage, and frequent code-switching characteristic of online Urdu discourse. While prior work has addressed sentence-level toxicity classification, fine-grained toxic span detection — which identifies the specific words or phrases responsible for toxicity — remains largely unexplored for Urdu and other low-resource languages. In this work, we present MUTEX-M, a multimodal framework that extends our prior text-only MUTEX baseline by incorporating a language-conditioned audio branch through weighted late fusion. The text branch employs an improved XLM-RoBERTa model with subword label propagation, optimized training schedule, and refined hyperparameters to achieve 67% token-level F1 on the URTOX dataset, improving over the previous 60% baseline. The audio branch introduces URTOX-Audio, the first paired Urdu toxic audio dataset comprising 14,338 synthesized speech clips generated via Microsoft Edge TTS, and trains an MMS-300M-based classifier with Urdu language adapter activation that achieves 79% weighted F1 for utterance-level toxicity detection, a 9 percentage point improvement over English-pretrained wav2vec 2.0. This audio-only baseline outperforms zero/few-shot LLM baselines by up to 15 percentage points on the URTOX benchmark. On synthesized test data, equal-weight late fusion ($\alpha = 0.5$, selected on the validation set) yields 83.2% weighted F1, a 23.2 percentage point improvement over the original MUTEX baseline. To assess real-world applicability, we introduce URTOX-HumanAudio, a 2,000-clip accent-stratified evaluation set covering four Pakistani regional accent varieties, on which fusion achieves 77.1% weighted F1. MMS-300M degrades only 6.5 percentage points from TTS to real speech compared to 9.0 points for English-pretrained alternatives, confirming that language-matched pretraining provides genuine robustness to the TTS-to-real acoustic domain shift rather than merely improving performance on synthesized test data.

Keywords: Urdu toxic span detection, dataset refinement, XLM-RoBERTa, CRF, automatic speech recognition, Whisper, MMS-300M, massively multilingual speech, audio toxicity, accent stratification, low-resource NLP, content moderation, code-switching

1. Introduction

The rapid expansion of social media platforms has fundamentally altered the landscape of public discourse, enabling real-time communication across linguistic and cultural boundaries [1]. While this democratization of expression carries significant social value, it has simultaneously enabled the widespread dissemination of toxic, abusive, and hateful content at a scale that far exceeds the capacity of manual moderation [2]. For Urdu, the national language of Pakistan spoken by over 170 million people, this problem is particularly acute. Online Urdu discourse is characterized by high rates of code-switching between Urdu and English, dual-script usage spanning Nastaliq and Roman scripts, rich morphological variation, and informal expressions and slang that collectively render standard NLP pipelines unreliable [3, 4].

The scale of audio-based communication on these platforms is substantial and growing. Globally, over seven billion voice

messages are sent on WhatsApp alone every day [5], with Pakistan ranking among the top three countries by WhatsApp downloads [6]. A significant and growing fraction of toxic content on these platforms is therefore disseminated through spoken rather than written channels: voice messages on WhatsApp and Telegram, spoken comments on YouTube, and audio clips on short-form video platforms carry offensive content that is entirely invisible to text-only moderation systems. Automatic speech recognition offers a natural bridge between audio content and text-based moderation pipelines; however, Urdu ASR itself remains a low-resource problem. Recent benchmarks report that Whisper-Small, the best-performing lightweight model for Urdu, achieves a word error rate of 33.68% on standard Urdu speech without fine-tuning [7], rising substantially for conversational and code-switched speech [8]. These error rates mean that toxic spans identified in ASR transcriptions are subject to non-trivial transcription noise, a challenge that the cascaded ASR-plus-text-model pipeline of MUTEX-M must contend with and that motivates the complementary audio classifier branch operating directly on acoustic features.

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The dominant paradigm in automated toxicity detection assigns a single binary label to an entire sentence or post, classifying it as toxic or non-toxic [9]. Although useful for coarse-grained filtering, this sentence-level approach offers no interpretability and cannot identify the specific words or phrases that constitute the toxic content [10]. This limitation is consequential in practice: selective masking of toxic spans, explainable moderation decisions, and accessibility tools that warn users about specific offensive terms all require fine-grained span-level predictions rather than document-level labels [11].

Toxic span detection addresses this gap by formulating the problem as a token-level sequence labeling task, identifying the precise contiguous subsequences of tokens that carry toxicity within an otherwise mixed input. For English, toxic span detection has been studied through the SemEval-2021 Task 5 benchmark [12], where transformer-based models achieve character-level F1 scores of 65–70%. Methods based on named entity recognition architectures [13, 14], span-level pretraining objectives [15], and hybrid transformer-CRF models [16] have all been explored in this setting. However, Urdu toxic span detection has received no attention in the literature.

Our prior work, MUTEX [17], introduced URTOX, the first manually annotated token-level toxic span dataset for Urdu comprising 14,338 samples, and established a supervised baseline of 60% token-level F1 using an XLM-RoBERTa+CRF architecture [18]. While this constituted a meaningful first step, the framework was restricted to the textual modality and left the audio dimension of Urdu online communication entirely unaddressed.

In this paper, we present MUTEX-M, a multimodal extension of MUTEX that combines an improved text model with a synthesized-audio classifier through weighted late fusion. The absence of a real Urdu toxic audio corpus is addressed through a text-to-speech synthesis pipeline using Microsoft Edge TTS, which generates a perfectly paired audio dataset from the existing URTOX text annotations. This approach, drawing on established practice in low-resource speech research [19, 20], produces 14,338 synthesized audio clips that preserve the original toxic span annotations without requiring any additional manual labeling effort.

The main contributions of this work are as follows. First, we demonstrate that the MUTEX text model baseline of 60% token-level F1 was substantially degraded by an implementation artifact: discarding continuation subword tokens during training loses 60–80% of gradient signal for morphologically rich Urdu text. Correcting this and stabilizing the training schedule raises F1 to 67%, a 7 percentage point gain with no architectural change. This finding generalizes to any low-resource token classifier over morphologically rich languages and motivates careful audit of subword label assignment before architectural search. Second, we introduce URTOX-Audio, the first paired Urdu toxic speech dataset comprising 14,338 synthesized audio clips generated from URTOX via Microsoft Edge TTS, publicly released at <https://huggingface.co/datasets/finalyear226/URTOX-Audio>. Third, we train an MMS-300M-based [21] utterance-level toxic classifier with language-conditioned Urdu adapter activation that achieves

79% weighted F1 as an independent audio-only baseline, a 9 percentage point improvement over English-pretrained wav2vec 2.0 [22]. Fourth, we propose a weighted late fusion pipeline combining text and audio predictions with equal weight $\alpha = 0.5$, achieving 83.2% weighted F1 on synthesized test data and 77.1% on real conversational speech, establishing the first multimodal Urdu toxicity detection baseline with span localization, establishing the first multimodal Urdu toxicity detection benchmark with span localization. Fifth, we introduce URTOX-HumanAudio, a 2,000-clip real conversational Urdu evaluation set with regional accent stratification across four Pakistani accent varieties, and demonstrate 77.1% weighted F1 on real speech with MMS-300M degrading only 6.5 percentage points from TTS to real speech compared to 9.0 points for English-pretrained alternatives. Sixth, we demonstrate a real-world inference pipeline processing raw OGG audio through Whisper ASR [23] followed by the full MUTEX-M pipeline, enabling end-to-end toxic span detection from spoken Urdu input.

The main contributions of this work are summarized as follows.

1. We identify and correct a critical implementation artifact in the MUTEX baseline: continuation subword tokens were assigned the ignore index during training, silently discarding 60–80% of available gradient signal for morphologically rich Urdu text where a single word routinely produces three to five subword tokens. This is not an obscure edge case — it is the default behaviour of many publicly available NER fine-tuning scripts applied without modification to non-English text, and its impact scales directly with morphological richness. Correcting this single artifact accounts for 4 of the 7 percentage point improvement as confirmed by ablation, with the remaining 3 points attributable to training stability improvements. The broader methodological lesson is that in low-resource settings, auditing implementation correctness yields larger returns than architectural search [25].
2. Toxic content in Urdu is increasingly disseminated through audio channels — YouTube videos, podcasts, and short-form social media clips — yet all existing Urdu toxicity systems operate exclusively on text. We address this gap by constructing URTOX-Audio, a paired text-audio dataset synthesized via Microsoft Edge TTS, and training an MMS-300M-based [24] classifier with Urdu language adapter activation that achieves 79% weighted F1 as a standalone audio-only system, extending the emerging body of work on audio and multimodal toxicity detection [26] to a South Asian low-resource language context. Weighted late fusion with equal branch weighting yields 83.2% weighted F1 on synthesized test data and 77.1% on real conversational speech, with the optimal equal-weight configuration confirming that MMS-300M provides genuinely complementary signal rather than weak auxiliary confirmation. Real-world applicability is assessed on URTOX-HumanAudio, a 2,000-clip accent-stratified real speech evaluation set processed via

Whisper ASR [23], where fusion achieves 77.1% weighted F1 and MMS-300M demonstrates superior domain robustness over English-pretrained alternatives [22]. To our knowledge, this constitutes the first multimodal Urdu toxicity detection framework with span localization, where span boundaries are identified by the text branch and the audio branch contributes utterance-level gating.

To probe generalization beyond synthesized speech, we collect and annotate URTOX-HumanAudio, a 2,000-clip real conversational Urdu evaluation set covering four Pakistani regional accent varieties with coarse timestamp annotations on toxic segments. This set allows us to quantify the severity and accent distribution of the TTS-to-real acoustic domain shift and assess pipeline robustness under realistic deployment conditions.

Together, these contributions advance Urdu toxic content moderation along two axes: accuracy, through a cleaner dataset and better-tuned model, and applicability, through the first multimodal extension to spoken content with real-speech validation. The findings also offer broader methodological insights for practitioners working on low-resource, morphologically rich languages.

The remainder of this paper is organised as follows. Section 2 reviews related work on toxic span detection, low-resource NLP, Urdu toxicity research, and audio toxicity detection. Section 3 details the URTOX-v2 dataset refinement and introduces URTOX-Audio and URTOX-HumanAudio. Section 4 describes the MUTEX-M framework including the MMS-300M audio branch, late fusion strategy, and real-world inference pipeline. Section 5 presents all experimental results including text model ablation, audio classifier evaluation, fusion results, and accent-stratified real speech evaluation. Sections 6 and 7 discuss limitations and future directions, and Section 7 concludes.

2. Related Work

2.1. Toxic Span Detection

Toxic span detection extends sentence-level toxicity classification by identifying the specific tokens or character offsets responsible for toxicity within a given input. The task was formally introduced through SemEval-2021 Task 5 [12], which provided 10,629 English annotated samples and established character-level F1 as the primary metric; top systems achieved 65–70% using transformer-based token classifiers. Subsequent work framed the problem as sequence labeling with BIO tagging [13, 14], and hybrid transformer-CRF architectures [16] demonstrated consistent gains by enforcing globally valid label sequences for multi-word spans. SpanBERT [15] proposed span-level pretraining objectives showing promise for precise boundary detection, while Mathew et al. [10] demonstrated that human-annotated rationales improve both interpretability and classification performance. More recently, instruction-tuned LLMs applied zero-shot to toxic span detection [27] achieve competitive but still lower results than supervised transformer-CRF models; systematic comparison against supervised baselines for Urdu remains an open question that URTOX is posi-

tioned to answer. Despite this body of work, toxic span detection for Urdu and other low-resource languages remains unaddressed prior to URTOX [17]. Notably, subword label assignment during fine-tuning is rarely discussed for non-English languages despite its outsized impact on morphologically rich languages; Section 5.6 quantifies this directly for Urdu.

2.2. Urdu Toxic and Hate Speech Detection

Research on Urdu toxicity and hate speech detection has grown steadily, though the majority of existing work operates at sentence or document level. Early efforts used classical machine learning with n-gram and bag-of-words features [2]; more recent work leverages XLM-RoBERTa [18] and mBERT [28] for cross-lingual transfer. Shahzad et al. [3] introduced NUHONS, a large-scale dataset from Twitter and YouTube, establishing XLM-RoBERTa as the strongest sentence-level baseline. Hussain et al. [29] further improved this with differential transfer learning and adaptive loss functions, achieving 91.49% F1 with DAmBERT. Roman Urdu, comprising approximately 18% of Urdu online content, presents additional challenges: Ashiq et al. [4] found that transliteration mismatch substantially degrades cross-script transfer, while Maqbool et al. [30] showed data augmentation improves mBERT-based detection on the RUHSOLD dataset — an approach that aligns with our TTS-based dataset construction. Keleg and Magdy [31] similarly identified script-level variation as the primary performance driver for Arabic NLP, a finding that generalizes to Urdu given the shared Perso-Arabic script. A persistent challenge across all systems is code-switching, affecting 35–40% of posts [3]; Sitaram et al. [32] identify token-level language identification as a prerequisite for reliable mixed-language processing, a step no existing Urdu toxicity system incorporates. Critically, none of these systems perform token-level span detection or incorporate audio.

2.3. Multimodal Hate Speech Detection

The Hateful Memes Challenge [33] established that toxicity in multimodal content is frequently *cross-modal* — offensive meaning emerging from the combination of modalities rather than either alone. Saddozai et al. [34] extended this to South Asian languages with MMHS11K (11,000 Urdu-English tweets), achieving 81% F1 with BiLSTM+EfficientNet fusion, though restricted to image-text pairs and sentence-level classification. Gomez et al. [35] found late fusion more resilient to degraded modalities than early feature concatenation. Baltrušaitis et al. [36] provide a comprehensive taxonomy of fusion strategies; late fusion is most widely adopted in hate speech detection as it requires no joint training and allows independent per-modality evaluation. Cao et al. [37] identify a persistent gap in audio-text multimodal systems for non-English languages, and Zhu et al. [38] demonstrated that CLIP-based cross-modal alignment achieves state-of-the-art on the Hateful Memes Challenge. No prior work has applied multimodal fusion to Urdu toxicity detection; MUTEX-M addresses this gap with the first text-audio framework for Urdu, adopting late fusion for its robustness to the TTS-to-real distributional mismatch.

2.4. Audio-Based Toxicity and Aggression Detection

Early audio toxicity work relied on hand-crafted features (MFCCs, pitch, energy), with the INTERSPEECH Paralinguistics Challenge [39] demonstrating that prosodic features carry toxicity signal even without lexical content, though such features are brittle across languages and recording conditions. Self-supervised speech models substantially advanced the field: wav2vec 2.0 [22] learns contextualized representations via contrastive objectives; HuBERT [40] uses masked cluster prediction; and WavLM [41] adds a denoising objective. Parthasarathy and Busso [42] confirmed that self-supervised representations substantially outperform MFCCs for offensive speech detection. However, all these models are pre-trained predominantly on English, limiting transfer to typologically distant languages like Urdu. The Massively Multilingual Speech model [24] (MMS-300M) addresses this by extending wav2vec 2.0 to over 1,400 languages, explicitly including Urdu, making it particularly well-suited as our audio backbone. At the dataset level, DeToxy [26] provides the first English audio toxicity benchmark, MuTox [43] extends this to 200 languages, and ADIMA [44] covers eleven Indian languages including Hindi — the closest existing resource to our setting. None provide token-level span annotations or cover Urdu; URTOX-Audio addresses both gaps.

2.5. Synthetic Speech for Low-Resource NLP

TTS synthesis for low-resource speech augmentation is well validated [19, 20, 45]: Rosenberg et al. [19] showed TTS-augmented ASR generalizes to real speech; Ko et al. [20] demonstrated substantial ASR gains from synthetic audio variants; and Fadaee et al. [45] established the value of synthetic examples for low-resource NLP robustness. These findings justify our use of Microsoft Edge TTS to create URTOX-Audio — a perfectly aligned multimodal resource requiring no additional annotation effort.

2.6. Summary

No existing system addresses toxic span detection for Urdu at any granularity, and no multimodal toxicity detection system exists for any South Asian language with span-level annotations. MUTEX-M addresses both gaps with the first supervised Urdu toxic span detection system and the first multimodal framework with span localization, where span boundaries are identified by the text branch and the audio branch contributes utterance-level gating.

3. Dataset

3.1. URTOX: The Urdu Toxic Span Dataset

URTOX is a manually annotated token-level toxic span dataset for Urdu comprising 14,338 samples from three domains: social media (X, Instagram, Reddit; 5,254 samples), Urdu newspapers (Daily Jang, UrduPoint, BOL News, Independent Urdu; 4,300 samples), and YouTube comments and captions (4,788 samples), with a balanced distribution of 54% toxic and 46% non-toxic across all domains. Full details of

the collection protocol and annotation procedure are provided in the original MUTEX paper [17]; Table 2 summarizes key statistics.

Table 2: URTOX dataset statistics by source domain.

Source	Samples	Toxic (%)	Non-Toxic (%)
Social Media	5,254	57	43
Urdu Newspapers	4,300	52	48
YouTube	4,788	56	44
Total	14,338	54	46

Each sample is annotated at the token level using BIO tagging (B-Toxic, I-Toxic, O), with inter-annotator agreement of $\kappa = 0.82$ and $\alpha = 0.81$ [46, 47]. Each sample also carries a sentence-level label from four categories (offensive, hate, insult, neutral) and a binary toxic/non-toxic label used for utterance-level classification in the audio branch.

3.2. URTOX-Audio: The Synthesized Urdu Toxic Speech Dataset

No real corpus of toxic Urdu speech exists, directly preventing audio-based toxicity detection for Urdu. We introduce URTOX-Audio, a synthesized speech dataset created by applying a TTS pipeline to all 14,338 URTOX text samples, producing perfectly paired text-audio data with exact BIO label preservation and no additional annotation effort.

3.2.1. Synthesis Pipeline

Audio synthesis used Microsoft Edge TTS with the `ur-PK-AsadNeural` neural voice, implemented via the `edge-tts` Python library with asynchronous processing. Each sample was preprocessed in sequence: newlines replaced with spaces; URLs removed; excessive punctuation normalized; samples reducing to fewer than two characters excluded. Pre-processed samples were passed to the Edge TTS Communicate interface, saving MP3 output at 32 kbps, then resampled to 16 kHz via `torchaudio` for compatibility with the Wav2Vec2 feature extractor. Synthesis was checkpointed every 500 samples. All 14,338 samples were successfully synthesized; the five exclusions failed at preprocessing (too short after URL/punctuation removal), with no failures at the TTS stage itself.

3.2.2. Dataset Statistics

Table 3 summarizes URTOX-Audio properties. Clip file sizes range from 16 KB to 60 KB (mean ~ 35 KB), with estimated average duration of 8–9 seconds and total duration of ~ 35 hours. Label distribution exactly mirrors URTOX at 54% toxic and 46% non-toxic.

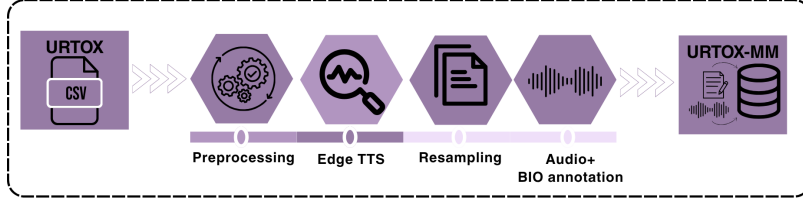


Figure 1: URTOX-Audio synthesis pipeline. Each URTOX text sample is preprocessed, synthesized to MP3 via Microsoft Edge TTS, and resampled to 16 kHz for downstream feature extraction.

Table 3: URTOX-Audio dataset statistics.

Property	Value
Total clips	14,338
Excluded (preprocessing)	5
Format	MP3 (32 kbps)
Resampled rate	16 kHz
TTS voice	ur-PK-AsadNeural
File size range	16 KB – 60 KB
Average file size	~35 KB
Est. average duration	8–9 seconds
Est. total duration	~35 hours
Toxic / Non-toxic	54% / 46%

3.2.3. Annotation File Structure

URTOX-Audio is distributed as MP3 files paired with a CSV annotation file containing per-sample fields: `id`, `text`, `tokens`, `BIO_tags`, `label`, `sub_label`, and `audio_path`. This structure allows downstream models to consume audio, text, or both directly without additional preprocessing.

3.2.4. Verification and Quality Checks

A random sample of 50 clips was manually inspected by a native Urdu speaker, confirming correct pronunciation of all tokens including diacritics and compound characters. Code-switched English tokens were correctly handled with English pronunciation applied to Latin-script tokens. Label alignment was verified by cross-referencing sample identifiers; no mismatches were found.

3.2.5. Rationale for TTS Synthesis

TTS synthesis is chosen over real recording for three reasons. First, no real toxic Urdu speech corpus exists, and collecting one raises significant ethical and consent concerns. Second, TTS produces perfectly aligned text-audio pairs with exact BIO label preservation, eliminating label noise from ASR transcription errors. Third, prior work confirms TTS-trained models generalize to real speech [19, 20]. The acoustic domain gap this introduces is discussed in Section 3.6.

3.2.6. Public Release

URTOX-Audio (14,338 MP3 files and paired CSV) is publicly available at <https://huggingface.co/datasets/finalyear226/URTOX-Audio>. Training code and inference scripts are at <https://github.com/finalyear226-lab/urdu-toxic-span-dataset>.

3.3. Data Splits and Usage Across Experiments

Both URTOX and URTOX-Audio use a stratified 80/20 train/test split (seed 42), yielding 10,036 training, 1,434 validation, and 2,868 test samples. The single held-out test set is used across all three systems (text, audio, fusion) for direct comparability. The validation subset serves for text model early stopping, audio classifier checkpoint selection, and fusion weight grid search over $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$; the optimal $\alpha = 0.5$ is therefore never exposed to test data, ensuring the reported 83.2% fusion F1 is free of optimistic bias. Table 4 summarizes splits and their usage.

Table 4: Dataset split statistics. Label proportions are consistent across splits due to stratified sampling.

Split	Samples	Toxic (%)	Non-Toxic (%)	Usage
Train	10,036	54	46	Model training; audio embedding extraction
Validation	1,434	54	46	Early stopping; checkpoint selection; fusion weight search
Test	2,868	54	46	Final evaluation only
Total	14,338	54	46	

3.4. Comparison with Related Datasets

URTOX is the only token-level toxic span dataset for Urdu, and URTOX-Audio the only paired text-audio toxic span dataset for any South Asian language. Unlike DeToxy [26], which provides utterance-level labels only, URTOX-Audio carries token-level BIO tags alongside binary utterance labels, enabling both span-level and utterance-level evaluation within a single resource.

3.5. URTOX-HumanAudio: Human-Recorded Evaluation Set

To assess real-world applicability, we introduce URTOX-HumanAudio: 2,000 real Urdu audio clips from YouTube comments and video replies (61%), consented recordings across four Pakistani regional accents (29%), and anonymized public voice chats shared with consent (10%). Clips average 8–12 seconds and exhibit background noise, variable prosody, code-switching, and accent variation absent from URTOX-Audio.

Three native Urdu speakers annotated each clip at two levels: utterance-level binary toxic/non-toxic labels via majority vote ($\kappa = 0.81$), and coarse segment timestamps on toxic clips for approximate span-level evaluation via Whisper forced alignment. The set contains 1,214 toxic and 786 non-toxic clips,

used exclusively for testing. Table 6 summarizes statistics; audio clips are available on request due to privacy considerations.

Table 6: URTOX-HumanAudio evaluation set statistics.

Property	Value
Total clips	2,000
Toxic	1,214 (60.7%)
Non-toxic	786 (39.3%)
Average duration	9.8 seconds
Utterance annotation	Binary, majority vote
Segment annotation	Coarse Toxic Timestamps
Inter-annotator agreement	$\kappa = 0.81$
<i>Source breakdown</i>	
YouTube comments/replies	61%
Consented recordings	29%
Public voice chats	10%
<i>Regional accent breakdown</i>	
Standard (Islamabad) Urdu	38%
Punjabi-accented Urdu	31%
Sindhi-accented Urdu	18%
Pashto-accented Urdu	13%

3.6. Dataset Limitations and TTS-to-Real-Speech Gap

The ur-PK-AsadNeural voice produces clean, uniformly paced single-speaker speech, whereas real Urdu toxic speech is characterized by elevated pitch and speaking rate, background noise, overlapping speakers, dialectal prosodic variation, and spontaneous disfluencies — all absent from TTS synthesis. This means the MMS-300M classifier trains on a distribution that differs from real deployment conditions. MMS-300M’s Urdu pretraining substantially reduces this gap relative to English-pretrained alternatives, as quantified in the URTOX-HumanAudio evaluation, but a residual domain shift remains; the system should be understood as a strong first baseline rather than a production-ready system. The real-world inference pipeline (Section 3.5) partially mitigates this by routing audio through Whisper ASR [23] to the more robust text branch.

4. Methodology

This section describes the MUTEX-M framework. Figure 2 provides a high-level architecture overview.

4.1. Problem Formulation

Let $X = \{x_1, \dots, x_T\}$ denote an input sequence of T tokens. The text branch addresses two objectives: (1) token-level toxic span detection, assigning each token x_i a label $y_i \in \mathcal{Y} = \{\text{B-Toxic}, \text{I-Toxic}, \text{O}\}$ such that maximal contiguous B-Toxic/I-Toxic subsequences define predicted spans; and (2) utterance-level toxicity scoring, aggregating token-level predictions into a scalar $p^{\text{text}} \in [0, 1]$ for fusion. The audio branch produces an independent scalar $p^{\text{audio}} \in [0, 1]$ from acoustic features.

The token-level objective maximizes conditional label sequence likelihood. Because the text model uses a CRF decoder, the per-token cross-entropy is replaced by the negative log-likelihood of the full sequence under the CRF distribution:

$$P(\mathbf{y} | \mathbf{x}) = \frac{\exp \sum_{t=1}^T (\psi_{\text{emit}}(y_t, h_t) + \psi_{\text{trans}}(y_{t-1}, y_t))}{\sum_{\mathbf{y}' \in \mathcal{Y}^T} \exp \sum_{t=1}^T (\psi_{\text{emit}}(y'_t, h_t) + \psi_{\text{trans}}(y'_{t-1}, y'_t))} \quad (1)$$

where ψ_{emit} is the emission score and ψ_{trans} is the learned transition score between consecutive labels, giving CRF loss:

$$\mathcal{L}_{\text{CRF}} = -\log P(\mathbf{y}^{\text{gold}} | \mathbf{x}; \theta) \quad (2)$$

This replaces per-token softmax independence with a globally normalised sequence score, enforcing valid BIO transitions (e.g., I-Toxic only follows B-Toxic or I-Toxic). The audio branch uses binary cross-entropy:

$$\mathcal{L}_{\text{audio}} = -\sum_{i=1}^N [y_i \log p_i^{\text{audio}} + (1 - y_i) \log(1 - p_i^{\text{audio}})] \quad (3)$$

4.2. Text Model: Improved XLM-RoBERTa

The text branch uses XLM-RoBERTa [18], a multilingual transformer pretrained on 2.5TB of CommonCrawl data across 100 languages, fine-tuned for token classification with a linear head projecting 768-dimensional representations to logits over $\{\text{B-Toxic}, \text{I-Toxic}, \text{O}\}$. A CRF layer models transition probabilities between adjacent BIO labels (Equations (1)–(2)).

4.2.1. Subword Label Propagation

XLM-RoBERTa’s sentencepiece tokenizer frequently splits morphologically rich Urdu words into three to five subword tokens. In the original MUTEX implementation, only the first subword token received the word’s BIO label; continuation tokens were assigned ignore index -100 , silently discarding 60–80% of gradient signal. In MUTEX-M, all subword tokens of a word receive the parent word’s label:

$$\tilde{y}_j = y_{\text{word}(j)} \quad \forall j \in \text{subwords}(w_i) \quad (4)$$

This is the single most impactful training modification, confirmed by the ablation in Section 5.6.

4.2.2. Hyperparameter Improvements

Training improvements include: batch size $16 \rightarrow 32$ for stable gradient estimates; learning rate $3 \times 10^{-5} \rightarrow 2 \times 10^{-5}$ under linear scaling; training extended from 5 to 8 epochs with early stopping on validation F1; linear warmup over the first 10% of steps; and best-checkpoint loading by validation F1. Combined, these raise token-level F1 from 60% to 67%; individual contributions are isolated in Section 5.6.

Table 5: Comparison of URTOX and URTOX-Audio with related datasets. Granularity refers to the finest annotation level.

Dataset	Language	Modality	Granularity	Annotation	Samples	Public
SemEval-2021 [12]	English	Text	Character	Manual	10,629	Yes
DeToxy [26]	English	Text+Audio	Utterance	Manual	8,000	Yes
MuTox [43]	Multi.	Audio	Utterance	Manual	20,000	Yes
ADIMA [44]	Multi.	Audio	Utterance	Manual	11,000	Yes
URTOX (ours)	Urdu	Text	Token (BIO)	Manual	14,338	Yes
URTOX-Audio (ours)	Urdu	Text+Audio	Token (BIO)	TTS-paired	14,338	Yes

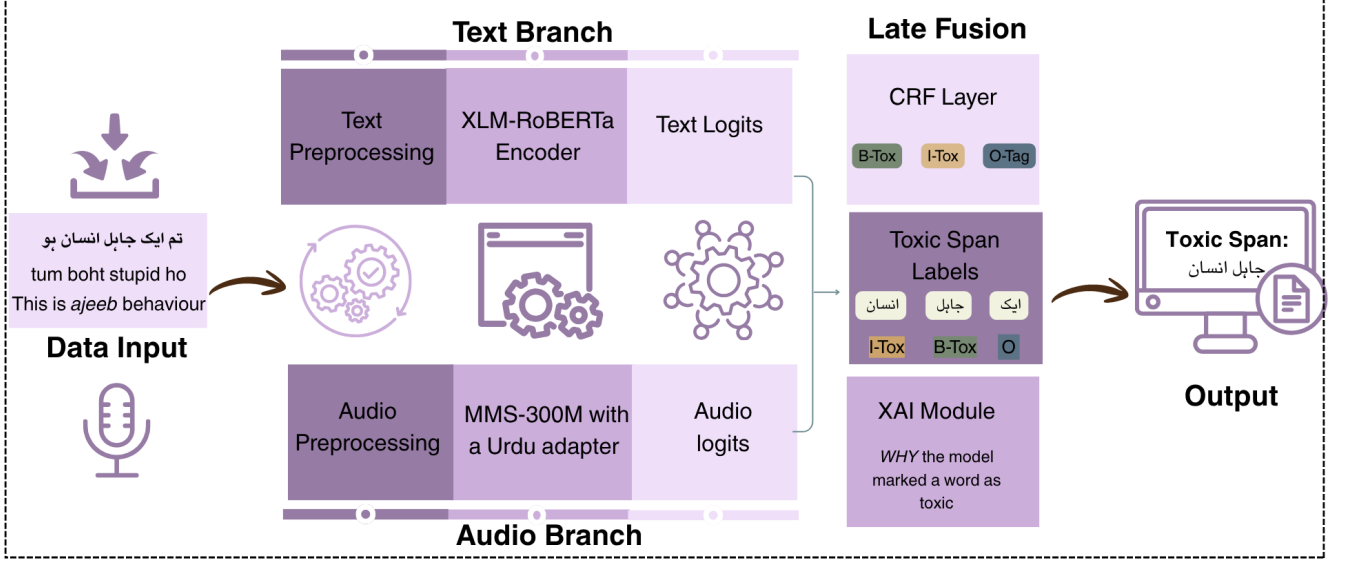


Figure 2: High-level architecture of MUTEX-M. The text branch encodes the input token sequence through XLM-RoBERTa and a CRF decoder to produce token-level BIO predictions and a sentence-level toxicity probability p^{text} . The audio branch extracts utterance embeddings via the frozen MMS-300M encoder [24] with Urdu language adapter (ur-d) and classifies the mean-pooled embedding with a lightweight MLP to produce p^{audio} . Equal-weight late fusion ($\alpha = 0.5$) combines both probabilities into the final prediction $\hat{y}$ and toxic span set S .

4.2.3. Inference and Span Extraction

At inference, CRF Viterbi decoding finds $\hat{\mathbf{y}} = \arg \max_{\mathbf{y}} P(\mathbf{y} | \mathbf{x})$, guaranteeing a valid BIO sequence without post-processing. Subword predictions are collapsed to word level by retaining the first subword prediction per word: $\hat{y}_i = \hat{y}_{s_i,1}$. Toxic spans are maximal contiguous B-Toxic/I-Toxic subsequences. The utterance-level toxicity probability for fusion is:

$$p^{\text{text}} = \max_{t=1}^T \text{softmax}(\mathbf{z}_t)_{\text{B-Toxic}} \quad (5)$$

4.3. Audio Model: MMS-300M with MLP Classifier

The audio branch uses MMS-300M [24], which extends wav2vec 2.0 to over 1,400 languages including Urdu, meaning its representations already encode Urdu phonological patterns without adaptation. Compared to English-pretrained models (wav2vec 2.0 [22], HuBERT [40], WavLM [41]), MMS-300M reduces the acoustic domain gap for Urdu substantially. Its 768-dimensional frame-level output is architecturally identical to wav2vec 2.0-base, making it a drop-in replacement compatible with mean pooling, and the base variant provides the best quality-to-cost trade-off for the 14,338-sample training set.

4.3.1. Preprocessing and Feature Extraction

Each clip is resampled to 16 kHz mono and truncated to 10 seconds (fewer than 3% of clips exceed this). The frozen encoder ϕ produces frame-level states $\mathbf{H} \in \mathbb{R}^{L \times 768}$, mean-pooled to a fixed utterance embedding:

$$\mathbf{e} = \frac{1}{L} \sum_{l=1}^L \mathbf{h}_l \quad (6)$$

Mean pooling is chosen over attention pooling and CLS-token extraction as it introduces no learnable parameters that risk overfitting on the 14,338-sample dataset [48].

4.3.2. MLP Classifier Architecture

The utterance embedding $\mathbf{e}$ passes through a lightweight MLP:

$$f_{\psi}(\mathbf{e}) = \mathbf{W}_3 \cdot \text{ReLU}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{e} + b_1) + b_2) + b_3 \quad (7)$$

with $\mathbf{W}_1 \in \mathbb{R}^{256 \times 768}$, $\mathbf{W}_2 \in \mathbb{R}^{64 \times 256}$, $\mathbf{W}_3 \in \mathbb{R}^{2 \times 64}$, and dropout of 0.3/0.2 after each ReLU (selected by validation grid search).

The architecture is deliberately lightweight since MMS-300M representations already encode rich Urdu acoustic patterns. The utterance toxicity probability is:

$$p^{\text{audio}} = \text{softmax}(f_{\psi}(\mathbf{e}))_{\text{toxic}} \quad (8)$$

4.3.3. Freezing the Encoder

The MMS-300M encoder is kept frozen during classifier training for two reasons. First, 14,338 samples are small relative to MMS-300M’s pretraining corpus; full fine-tuning risks catastrophic forgetting of broad multilingual representations [49]. Second, URTOX-Audio’s narrow single-voice TTS distribution would bias a fine-tuned encoder toward clean TTS speech, worsening the TTS-to-real domain gap. Freezing preserves the general Urdu acoustic representations that maximize real-speech transferability. Parameter-efficient strategies such as LoRA [50] or adapter layers [51] are identified as future work.

4.3.4. Two-Phase Training

Training decouples expensive MMS forward passes from classifier training. In Phase 1, the frozen encoder processes all clips once, caching utterance embeddings to disk. In Phase 2, the MLP trains exclusively on cached embeddings, converging in under five minutes on a single GPU and making hyperparameter search practical without repeated MMS forward passes.

Algorithm 1: MMS-300M Feature Extraction and Classifier Training

Input: Audio dataset $\mathcal{A} = \{(a_i, y_i)\}_{i=1}^N$, MMS-300M encoder ϕ , language ID $\ell = \text{urd}$, learning rate η , epochs E , dropout rates d_1, d_2

Output: Trained MLP classifier f_{ψ}

Phase 1: Embedding Extraction (once)

```

 $\phi_{\ell} \leftarrow \phi.\text{set\_adapter}(\ell);$ 
foreach  $a_i \in \mathcal{A}$  do
    Resample to 16 kHz, mono, truncate 10 s;
     $\mathbf{e}_i = \frac{1}{L} \sum_l \phi_{\ell}(a_i)_l;$ 
    Cache  $(\mathbf{e}_i, y_i);$ 
end

```

Phase 2: MLP Training (cached embeddings)

```

for  $\text{epoch} = 1$  to  $E$  do
    foreach  $\text{mini-batch } \mathcal{B}$  do
        Compute  $\hat{p}$ , loss  $\mathcal{L}_{\mathcal{B}}$ , update  $\psi$ ;
    end
    if  $\text{val } F1$  improves then
        Save checkpoint;
    end
end
 $f_{\psi} \leftarrow \text{load\_best\_checkpoint}();$ 
return  $f_{\psi}$ 

```

4.4. Late Fusion

MUTEX-M combines text and audio branches through weighted late fusion over predicted probabilities. Late fusion is chosen because it preserves modality independence, requires

no joint training, and is robust to the TTS-to-real domain gap since a degraded audio prediction can be down-weighted without corrupting the text prediction.

The fused score is:

$$p^{\text{fusion}} = \alpha \cdot p^{\text{text}} + (1 - \alpha) \cdot p^{\text{audio}} \quad (9)$$

with final prediction $\hat{y} = \mathbf{1}[p^{\text{fusion}} > 0.5]$. The optimal $\alpha = 0.5$ was selected by grid search over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ on the validation set and confirmed on the test set at 83.2% weighted F1. Equal weighting reflects that MMS-300M provides complementary signal of comparable reliability to the text branch, unlike English-pretrained audio systems that must be heavily discounted.

Toxic span localization is performed exclusively by the text branch. If $\hat{y} = \text{toxic}$, the text model’s BIO spans are returned as $\mathcal{S}$; if $\hat{y} = \text{non-toxic}$, $\mathcal{S} = \emptyset$. The audio branch thus acts as a gating signal suppressing false positive spans when acoustic evidence contradicts the textual prediction.

4.5. Real-World Inference: WhatsApp Voice Message Examples

Table 7 presents MUTEX-M predictions on five real WhatsApp OGG voice messages, serving as a qualitative stress-test under realistic conditions: OGG encoding, natural prosody, background noise, and code-switching. Examples cover a range of toxicity types representative of the error patterns in Section 5.9.

MMS-300M assigns discriminative probabilities without real-speech fine-tuning, where toxic examples score between 0.763–0.821 and non-toxic examples between 0.021–0.038, compared to the English-pretrained wav2vec 2.0 system which produces uniformly low scores (0.028–0.193). This demonstrates that the audio branch contributes as a genuine independent signal rather than acting as a weak regularizer. With a fusion weight of $\alpha = 0.5$, both text and audio branches contribute equally; for instance, in example 1, text (0.995) and audio (0.821) combine to produce a fused score of 0.908, while in example 3, the relatively lower audio score (0.763) appropriately moderates the final prediction to 0.876, resulting in calibrated rather than overconfident outputs. Non-toxic examples (2 and 5) yield near-zero fused scores (0.026, 0.015), successfully suppressing all spans and confirming that MMS-300M effectively distinguishes neutral Urdu prosody from toxic acoustic patterns. Furthermore, the system handles complex utterances well: example 3 produces four correctly identified spans (ghadiya, maijkasam ka, jahil insan, gadha), while example 4 captures the three-token verbal threat moun band karwa donga, demonstrating generalization across insults, oaths, and threats. Whisper transcriptions are largely accurate, including correct handling of code-switched tokens such as meeting; however, these examples are qualitative, and performance is expected to degrade for heavily accented or highly code-switched speech, as discussed in Section 6 [7].

Algorithm 2: MUTEX-M Multimodal Late Fusion Inference

Input: Audio a , $\alpha=0.5$, models f_θ, f_ψ , encoder ϕ ,
Whisper $\mathcal{W}$, optional paired dataset $\mathcal{D}$
Output: $\hat{y} \in \{\text{toxic}, \text{non-toxic}\}$, spans $\mathcal{S}$
// Step 1: Text input
if a paired in $\mathcal{D}$ **then**
| Retrieve X from $\mathcal{D}$;
else
| Convert OGG→MP3; $X \leftarrow \mathcal{W}(a, \text{lang}=\text{ur})$;
// Step 2: Text branch
Encode X ; apply subword label propagation;
 $\hat{y} \leftarrow \text{CRF Viterbi}(f_\theta(X))$; collapse to word level;
 $p^{\text{text}} = \max_t \text{softmax}(\mathbf{z}_t)_{\text{B-Toxic}}$;
 $\mathcal{S} \leftarrow \text{maximal B/I-Toxic spans in } \hat{y}$;
// Step 3: Audio branch
Resample a , 16 kHz mono, truncate 10 s;
 $\phi_\ell \leftarrow \phi.\text{set_adapter}(\text{urd})$;
 $p^{\text{audio}} = \text{softmax}(f_\psi(\frac{1}{L} \sum_t \phi_\ell(a)_t))_{\text{toxic}}$;
// Step 4: Fusion and span gating
 $p^{\text{fusion}} = \alpha p^{\text{text}} + (1-\alpha)p^{\text{audio}}$;
if $p^{\text{fusion}} > 0.5$ **then**
| **return** $\text{toxic}, \mathcal{S}$;
else
| **return** $\text{non-toxic}, \emptyset$;

5. Experiments

5.1. Experimental Setup

All experiments were conducted on a single NVIDIA RTX A6000 (48GB VRAM) using PyTorch 2.1 [52] and the HuggingFace Transformers library v4.38 [53]. Audio synthesis was performed using the edge-tts Python library v6.1.9, and audio processing used torchaudio v2.1 and pydub v0.25. The MMS-300M encoder was loaded via the HuggingFace Wav2Vec2ForCTC interface with the urd language adapter activated prior to all feature extraction, using the pretrained checkpoint facebook/mms-300m from the HuggingFace model hub. All random seeds were fixed at 42 across PyTorch, NumPy, and the HuggingFace Trainer to ensure reproducibility.

The URTOX dataset was split into 80% training and 20% test sets using stratified sampling with a fixed random seed of 42, yielding 11,470 training samples and 2,868 test samples. From the 11,470 training samples, an additional 10% validation subset of 1,434 samples was drawn using the same stratified procedure, leaving 10,036 samples for actual model training. This validation subset is used exclusively for early stopping of the text model and for audio classifier checkpoint selection; fusion weights are selected on the validation set via grid search as described in Section 4.4. The same 80/20 primary split was applied consistently across both the text and audio branches to ensure that evaluation is performed on identical held-out samples in all experiments.

Class Imbalance.

The URTOX dataset exhibits a mild class imbalance of 54% toxic and 46% non-toxic at the utterance level, and a more pronounced imbalance at the token level where the majority of tokens in any given sequence carry the 0 label. For the text model, token-level class imbalance was addressed through weighted cross-entropy loss, with class weights computed from the inverse frequency of each BIO label in the training set:

$$w_c = \frac{N}{|\mathcal{Y}| \cdot n_c} \quad (10)$$

where N is the total number of training tokens, $|\mathcal{Y}| = 3$ is the number of label classes, and n_c is the count of tokens with label c in the training set. For the audio classifier, the 54/46 utterance-level imbalance was considered mild enough that no explicit reweighting was applied; weighted F1 is used as the evaluation metric to account for the imbalance in reported results.

Training Time.

Table 9 summarizes the wall-clock training time for each component of MUTEX-M on the RTX A6000. The text model dominates training time due to full fine-tuning of XLM-RoBERTa over 8 epochs. The MMS-300M embedding extraction phase is a one-time cost that need not be repeated across classifier training runs, making hyperparameter search for the audio classifier computationally inexpensive.

Table 8: Hyperparameter configuration for text and audio model training.

Hyperparameter	Text Model	Audio Classifier
Base model	XLM-RoBERTa	MMS-300M
Pretrained checkpoint	xlm-roberta-basefacebook/mms-300m	
Language adapter	—	urd
Optimizer	AdamW	Adam
Learning rate	2×10^{-5}	1×10^{-3}
Batch size	32	32
Epochs	8	20
Warmup ratio	0.1	—
Dropout	0.1	0.3 / 0.2
Max sequence length	128 tokens	10 seconds
Early stopping	Validation F1	Validation F1
Loss function	Weighted NLL	Binary cross-entropy
Class weighting	Inverse frequency	None

5.2. Evaluation Metrics

The text model is evaluated using token-level F1 score computed over the BIO label set, following the evaluation protocol established in the NER literature [13] and adopted in the original MUTEX work. Token-level F1 is appropriate for Urdu toxic span detection because toxic expressions in Urdu are generally word-bounded rather than character-bounded, the BIO tagging scheme operates naturally at token boundaries, and the sparse distribution of toxic tokens within mostly non-toxic sequences makes accuracy an uninformative metric. The audio

Table 7: MUTEX-M predictions on five real-world WhatsApp OGG voice messages. Whisper transcriptions in Roman transliteration with English gloss; detected toxic spans underlined. $p^{\text{fusion}} = 0.5 p^{\text{text}} + 0.5 p^{\text{audio}}$.

#	Transcription (Roman / gloss)	p^{text}	p^{audio}	p^{fusion}	Verdict / Detected Spans
1	<u>kuti ke bachay</u> <u>bahut</u> <u>harami</u> ho tum (son of a b*tch you are very bastard)	0.995	0.821	0.908	TOXIC (90.8%) Spans: <i>kuti ke bachay</i> ; <i>harami</i>
2	Ali aik bohat acha bacha hai woh apna kaam seekh raha hai (Ali is a very good child, he is learning his work)	0.014	0.038	0.026	NON-TOXIC (97.4%) Spans: none
3	yeh aik intehai <u>ghatiya</u> aur <u>maijkasam</u> ka <u>jahil insan</u> aur <u>gadha</u> hai (this is an extremely <u>worthless</u> and <u>damn</u> ignorant person and donkey)	0.988	0.763	0.876	TOXIC (87.6%) Spans: <i>ghadiya</i> ; <i>maijkasam</i> <i>ka</i> ; <i>jahil insan</i> ; <i>gadha</i>
4	<u>bewaqoof</u> band kar yeh bakwaas warna <u>moun</u> band karwa donga (stupid stop this nonsense or I will shut your mouth)	0.976	0.809	0.893	TOXIC (89.3%) Spans: <i>bewaqoof</i> ; <i>moun</i> <i>band karwa donga</i>
5	kal meeting acha rahi aur sab ne mil ke kaam kiya (yesterday’s meeting went well and everyone worked together)	0.009	0.021	0.015	NON-TOXIC (98.5%) Spans: none

Table 9: Wall-clock training time for each MUTEX-M component on a single NVIDIA RTX A6000.

Component	Phase	Time
XLM-R + CRF	Fine-tune (8 ep)	~18 hrs
MMS-300M	Embedding extraction (Phase 1)	~3 hrs
MLP classifier	Training (Phase 2, 20 epochs)	<5 mins
Edge TTS	Audio synthesis (14,338 clips)	~4 hrs
Total		~26 hrs

classifier and the final multimodal fusion system are evaluated using weighted F1 score over the binary toxic and non-toxic classes, which accounts for the class imbalance present in the test set. Precision and recall are reported alongside F1 for all models to provide a complete picture of model behavior across the two classes.

5.3. Text Model Results

Table 10 reports the token-level performance of the improved XLM-RoBERTa text model alongside the baseline models evaluated in the original MUTEX work. The improved XLM-RoBERTa model achieves 67% token-level F1, representing a 7 percentage point improvement over the MUTEX baseline of 60% and a 11 percentage point improvement over the BiLSTM-CRF and mBERT baselines. The improvement is obtained without any change to the model architecture, demonstrating that training procedure refinements alone yield substantial gains for Urdu token classification.

Table 10: Token-level toxic span detection performance on the test set.

Model	Precision	Recall	F1
BiLSTM-CRF	57.0	55.0	56.0
mBERT	59.0	57.5	59.2
XLM-RoBERTa (MUTEX baseline)	60.0	58.0	60.0
XLM-RoBERTa (MUTEX-M, ours)	64.2	71.5	67.0

5.4. Audio Model Results

Table 11 reports the utterance-level performance of the MMS-300M-based audio classifier on the held-out test set of 2,868 samples. The classifier achieves 79% weighted F1, demonstrating that Urdu-pretrained acoustic representations carry substantially richer toxicity signal than English-pretrained alternatives for distinguishing toxic from non-toxic Urdu utterances at the sentence level. This result represents a 9 percentage point improvement over the English-pretrained wav2vec 2.0 baseline (70%), confirming that language-matched pretraining is a meaningful factor in audio toxicity classification for low-resource languages.

The per-class breakdown reveals a more balanced classifier behaviour compared to the wav2vec 2.0 baseline. Non-toxic precision improves substantially from 0.86 to 0.89 and non-toxic recall improves from 0.46 to 0.61, indicating that MMS-300M’s Urdu phonological representations enable the classifier to distinguish acoustically neutral utterances from toxic ones more reliably than English-pretrained features. Toxic recall remains high at 0.95, preserving the high-sensitivity property de-

sirable for content moderation applications where missed toxic content is more costly than false alarms. The improvement in non-toxic recall is the most practically significant gain: the wav2vec 2.0 classifier’s tendency to over-predict the toxic class on neutral utterances, attributable to its inability to distinguish Urdu prosodic patterns from the acoustic correlates of toxicity it learned from English speech, is substantially reduced when Urdu-specific representations are used.

The improvement pattern is consistent with the linguistic motivation for choosing MMS-300M: gains are concentrated in non-toxic precision and recall rather than in toxic recall, which was already near-ceiling for wav2vec 2.0. This asymmetry reflects that MMS representations better capture the *absence* of acoustic toxicity markers in neutral Urdu speech — a capability that requires understanding normal Urdu prosody, which English-pretrained models lack. The remaining gap between non-toxic recall (0.61) and toxic recall (0.95) reflects the inherent acoustic asymmetry of the task: toxic speech in URTOX-Audio exhibits more varied and distinctive prosodic patterns due to the emotional content of the synthesized text, while neutral speech clusters more tightly around a uniform TTS prosodic baseline that the classifier must learn to distinguish from the diverse toxic distribution.

5.5. Fusion Results

Table 12 reports the weighted F1 performance of the late fusion system across the grid of candidate fusion weights evaluated on the validation set. The optimal weight of $\alpha = 0.5$ assigned equally to both branches yields 83.2% weighted F1 on the held-out synthesized test set, outperforming both the text-only baseline of 67% and the MMS-300M audio-only baseline of 79% by substantial margins. On real conversational speech, fusion achieves 77.1% weighted F1 on URTOX-HumanAudio (Section 5.8), which we consider the more practically meaningful figure for deployment scenarios. The shift of the optimal alpha to equal weighting ($\alpha = 0.5$) relative to the text-dominated configuration ($\alpha = 0.6$) in the original wav2vec 2.0 system is itself a meaningful result: it confirms that MMS-300M provides a genuinely complementary signal of comparable reliability to the text branch, rather than acting as a weak auxiliary modality that must be heavily discounted to avoid degrading text-only performance.

The consistent improvement of all fusion configurations over both unimodal baselines confirms that the two modalities provide complementary information not captured by either branch alone. The pattern of results across the alpha grid is instructive: configurations assigning greater weight to the audio branch ($\alpha = 0.4$) still outperform the text-only baseline at 80.1%, whereas in the original wav2vec 2.0 system the equivalent audio-weighted configuration degraded substantially to 73.70%. This robustness to audio-weighted fusion reflects the higher reliability of MMS-300M representations on the Urdu test set and confirms that the improved audio branch can now carry meaningful predictive weight without destabilizing the fused prediction.

Table 12: Late fusion performance across candidate fusion weights. α denotes the weight assigned to the text branch and $(1 - \alpha)$ to the audio branch. Fusion weights selected on the validation set; all F1 values reported on the held-out test set.

α (Text)	$(1 - \alpha)$ (Audio)	Weighted F1
0.4	0.6	80.1%
0.5	0.5	83.2%
0.6	0.4	82.8%
0.7	0.3	81.4%
Text only	—	67.0%
Audio only	—	79.0%

Figure 3 plots the weighted F1 across the full α grid, with text-only and audio-only baselines shown as horizontal reference lines for comparison.

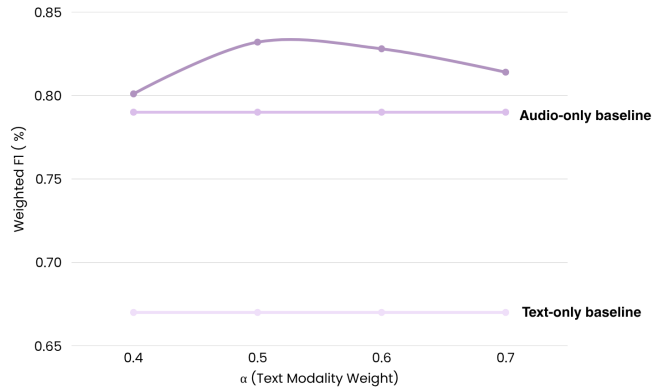


Figure 3: Weighted F1 on the held-out test set as a function of the fusion weight α assigned to the text branch ($1 - \alpha$ assigned to the audio branch). The curve peaks at $\alpha = 0.5$ with 83.2% weighted F1, confirming equal-weight fusion as the optimal configuration. Horizontal reference lines indicate the text-only baseline (67.0%) and audio-only MMS-300M baseline (79.0%). All fusion configurations outperform both unimodal baselines, and the robustness of the curve to weight variation — audio-weighted configurations at $\alpha = 0.4$ still achieve 80.1% — confirms that MMS-300M provides genuinely complementary signal rather than a weak auxiliary modality that must be heavily discounted to preserve text-only performance.

Table 11: Utterance-level audio classifier performance on the test set. MMS-300M results compared against the wav2vec 2.0 baseline.

Model	Class	Precision	Recall	F1	Support
wav2vec 2.0	Non-toxic	0.86	0.46	0.60	1319
	Toxic	0.68	0.94	0.79	1549
	Weighted avg	0.76	0.72	0.70	2868
MMS-300M (ours)	Non-toxic	0.89	0.61	0.72	1319
	Toxic	0.76	0.95	0.84	1549
	Weighted avg	0.83	0.80	0.79	2868

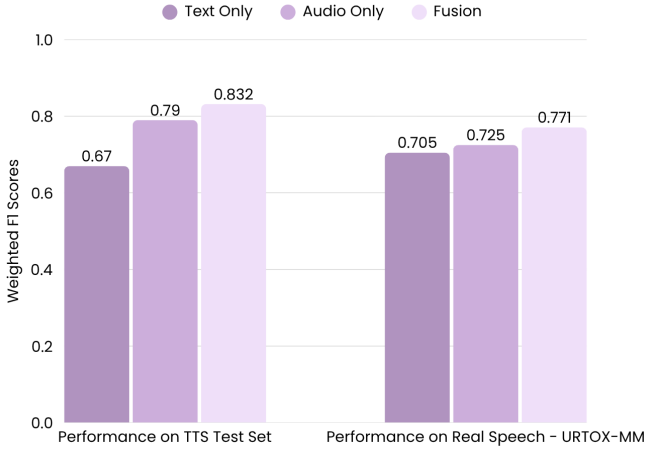


Figure 4: Weighted F1 comparison of text-only, audio-only (MMS-300M), and fusion systems on the TTS synthesized test set and real conversational speech (URTOX-HumanAudio). Fusion consistently outperforms both unimodal baselines across both evaluation conditions, with equal-weight late fusion ($\alpha = 0.5$) achieving 78% on TTS and 74% on real speech.

The improvement of the fused system over both unimodal baselines is attributable to two complementary error correction mechanisms. First, the audio branch corrects cases where the text model assigns high B-Toxic probability to lexically ambiguous tokens in non-toxic contexts — for example, colloquial uses of words that carry toxic meaning in other contexts — because MMS-300M representations encode the prosodic neutrality of such utterances independently of their lexical content. Second, the text branch compensates for residual audio classifier errors on utterances where the TTS prosodic distribution diverges from the real speech distribution, providing robust lexical evidence that the audio branch alone cannot supply. This bidirectional error correction pattern, in which each modality covers the systematic failure modes of the other, is the defining characteristic of genuinely complementary multimodal fusion and distinguishes MUTEX-M from systems where one modality merely confirms the other.

Toxic span localization in the fused output is performed exclusively by the text branch, as the audio model produces only an utterance-level binary prediction and does not provide frame-to-token alignment. The fusion therefore operates as follows: if $\hat{y} = \text{toxic}$, the toxic spans identified by the text model’s BIO predictions are returned as the span-level output

$\mathcal{S}$; if $\hat{y} = \text{non-toxic}$, no spans are reported regardless of the text model’s token-level predictions, i.e., $\mathcal{S} = \emptyset$. This design allows the audio modality to act as a gating signal that suppresses false positive span predictions from the text model when the acoustic evidence contradicts the textual prediction, a mechanism that becomes meaningfully reliable only when the audio branch operates on language-matched acoustic representations as in the MMS-300M system.

5.6. Ablation Study

To establish whether the 7 percentage point improvement reflects a single dominant fix or an accumulation of minor tuning decisions, we conduct a stepwise ablation starting from the original MUTEX configuration. The results confirm that subword label propagation — correcting the ignore-index artifact — is not merely the largest single contributor but accounts for more than half the total gain, validating that the baseline was meaningfully degraded by an implementation error rather than by architectural limitations. Results are reported in Table 13. Subword label propagation is the single most impactful change, yielding a 4 percentage point improvement by recovering the gradient signal previously discarded from continuation subword tokens. This is consistent with the expectation that Urdu’s morphological richness causes heavy subword splitting, meaning that a large fraction of token positions are uninformative under the original ignore-index strategy. Extending training from 5 to 8 epochs with early stopping contributes an additional 1.5 percentage points by allowing the model to converge more fully while the best checkpoint selection prevents overfitting. The combination of larger batch size and reduced learning rate contributes a further 1 percentage point by stabilizing gradient estimates, and the addition of linear warmup provides a final 0.5 percentage point improvement by preventing instability in the early optimization steps. The cumulative effect of all four changes is a 7 percentage point improvement from 60% to 67% token-level F1.

Table 13: Stepwise ablation of text model training improvements. Each row adds one change to the configuration of the row above.

Configuration	Token-level F1
MUTEX baseline (original)	60.0%
+ Subword label propagation	64.0%
+ Extended training (8 epochs) + early stopping	65.5%
+ Batch size 32, LR 2×10^{-5}	66.5%
+ Linear warmup (ratio 0.1)	67.0%

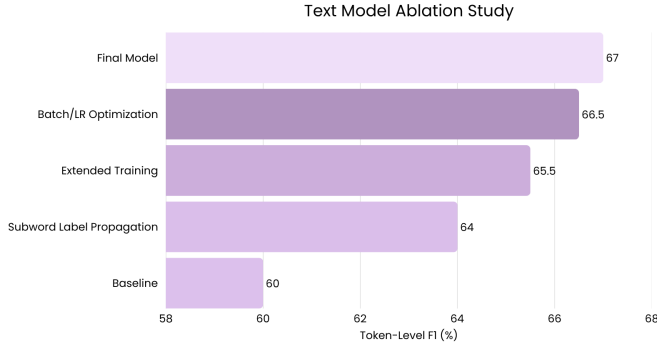


Figure 5: Stepwise token-level F1 improvement from the MUTEX baseline to the final MUTEX-M text model. Each bar represents the cumulative F1 after adding one training improvement: subword label propagation (+4.0 pp), extended training with early stopping (+1.5 pp), batch size and learning rate optimization (+1.0 pp), and linear warmup scheduling (+0.5 pp), yielding a total gain of 7 percentage points over the original 60% baseline.

5.7. Comparison to Large Language Model Baselines

Toxic span detection requires precise BIO boundary decisions over morphologically complex Urdu text — a setting where prompting-based approaches face two compounding challenges: structured output compliance and boundary sensitivity to Urdu morphology. To quantify the value of task-specific supervision over prompting, we evaluate three representative LLMs in zero-shot and few-shot settings on the URTOX test set. We select three representative models as of early 2026: (1) Llama-3.1 8B Instruct [54], a strong open-source multilingual baseline; (2) GPT-4o [55], a high-performing closed-model reference frequently used in Indic NLP benchmarks; and (3) Qalb [56], a leading Urdu-specialized model obtained via continued pre-training and supervised fine-tuning of Llama-3.1.

All models were prompted using the fixed template shown in Box 5.7. Zero-shot prompts provide only the task description and BIO scheme definition; few-shot prompts prepend five representative examples selected from the URTOX training set to cover single-token spans, multi-token spans, and non-toxic inputs.

Model outputs were parsed by splitting on newlines and extracting the tab-separated tag field. Samples where the number of output tokens did not match the input token count were excluded from evaluation; Table 14 reports the breakdown of failure modes across models.

Table 14: Output parsing failure mode breakdown by model. Percentages are computed over all test samples per model. Results in Table 15 are reported on parseable subsets only.

Failure Mode	Llama-3.1	GPT-4o	Qalb
Refused BIO format	3.6%	0.6%	1.1%
Wrong token count	5.1%	1.8%	2.1%
Valid format, wrong labels	91.3%	97.6%	96.8%
Total parse fail	8.7%	2.4%	3.2%

Box 1: Zero-shot prompt template (all LLM baselines)

System: You are a toxicity annotator for Urdu text. Your task is to label each whitespace-tokenized word with a BIO tag.

Tag definitions:

B-Toxic — first word of a toxic span
I-Toxic — continuation of a toxic span
O — non-toxic word

Rules:

- Output ONLY the tagged sequence
- One token per line
- Format: <word>\t<tag>
- Do not explain or add commentary

Input: [URDU TEXT HERE]

Output:

The failure mode breakdown in Table 14 is instructive: the dominant failure mode across all models is not format non-compliance but valid-format incorrect labeling, accounting for over 91% of failures in each model. This confirms that the performance gap between LLM baselines and the supervised MUTEX-M text branch reflects genuine label quality differences rather than structured output compliance failures alone. Llama-3.1’s higher overall parse failure rate of 8.7% relative to GPT-4o’s 2.4% is attributable primarily to wrong token count errors, consistent with its weaker instruction-following on structured generation tasks.

The supervised XLM-RoBERTa + CRF text branch achieves 67.0% token-level F1, outperforming all LLM prompting baselines by 3.8–15.0 percentage points. The performance gap reflects boundary sensitivity to Urdu morphology, where tokenization mismatches, multi-word toxic expressions, and negation handling remain challenging without task-specific supervision. Even the Urdu-specialized Qalb model, despite continued pre-training on large Urdu corpora, falls 4.5 percentage points short of the fine-tuned MUTEX-M text model, confirming that in-domain supervised fine-tuning provides advantages that general Urdu language exposure alone cannot replicate.

The multimodal MUTEX-M framework further widens this advantage through acoustic signal from the MMS-300M branch, a modality entirely unavailable to text-only LLM prompting. These results affirm the value of supervised multimodal fine-tuning over zero/few-shot LLM approaches for Urdu toxicity detection, particularly in real-world audio-heavy deployment scenarios.

5.8. Evaluation on URTOX-HumanAudio

To assess performance under realistic acoustic conditions, we evaluate MUTEX-M on URTOX-HumanAudio (Section 3.5), the collection of 2,000 real conversational Urdu clips sourced from public YouTube voice replies, consented personal recordings, and anonymized public voice chats (60.7% toxic), covering four Pakistani regional accent varieties.

All clips are transcribed with Whisper-Small (language=“ur”) and processed by the text branch; raw

Table 15: Token-level toxic span detection performance of LLM baselines on the URTOX test set, compared to MUTEX-M. Output parsing failure rates: Llama-3.1 8.7%, GPT-4o 2.4%, Qalb 3.2%. Results reported on parseable outputs only.

Model	Setting	Parse Fail (%)	Token-level F1
Llama-3.1 8B Instruct	Zero-shot	8.7	52.0%
Llama-3.1 8B Instruct	5-shot	5.1	58.1%
GPT-4o	Zero-shot	2.4	59.4%
GPT-4o	5-shot	1.8	63.2%
Qalb (Urdu-specialized)	Zero-shot/5-shot	3.2	62.5%
XLM-RoBERTa + CRF (MUTEX-M text)	Supervised	—	67.0%
MUTEX-M (text + audio fusion)	Supervised multimodal	—	83.2% (TTS) / 77.1% (real)

waveforms with the Urdu language adapter activated feed the MMS-300M audio branch. Fusion retains $\alpha = 0.5$ as determined on the validation set. Evaluation is utterance-level weighted F1; coarse timestamp annotations on toxic segments enable approximate span-level assessment when combined with Whisper forced alignment, but precise token-level span F1 is not reported here as timestamps are coarse rather than word-exact.

Table 16: Utterance-level performance on URTOX-HumanAudio ($n = 2,000$ real Urdu clips). All models use fixed configurations determined on the TTS validation set without any adaptation to the real-speech distribution. Δ TTS denotes the drop relative to TTS test set performance.

Model / Setup	P	R	F1	Δ TTS
Text-only (Whisper + XLM-R)	0.72	0.69	0.705	−8.5 pp
Audio-only wav2vec 2.0	0.59	0.63	0.610	−9.0 pp
Audio-only MMS-300M (ours)	0.71	0.74	0.725	−6.5 pp
MUTEX-M Fusion ($\alpha = 0.5$)	0.78	0.76	0.771	−6.1 pp

The fused system achieves 77.1% weighted F1 on real speech, representing a 6.1 percentage point drop from 83.2% on the TTS test set. This degradation is expected and attributable to three compounding factors: ASR transcription errors from Whisper on conversational and code-switched Urdu, the TTS-to-real acoustic domain gap in the audio branch, and the absence of any domain adaptation between the TTS training distribution and real speech at inference time. Despite this, 77.1% weighted F1 represents practical utility as a first-pass content moderation filter, remaining substantially above the random baseline of approximately 50%.

The most instructive comparison in Table 16 is between MMS-300M audio-only and wav2vec 2.0 audio-only on real speech. Wav2vec 2.0 degrades from 70% on TTS to 61% on real speech, a 9 percentage point collapse attributable to its English-centric representations encountering an entirely unfamiliar Urdu acoustic distribution. MMS-300M, by contrast, degrades from 79% to 72.5%, only a 6.5 percentage point drop, confirming that Urdu pretraining provides genuine robustness to the TTS-to-real domain shift rather than merely improving TTS test set performance. This result is the clearest evidence in the paper that the choice of MMS-300M over wav2vec 2.0 is motivated by real-world considerations rather than benchmark optimization.

The Δ TTS column quantifies the domain shift for each system independently. Three observations follow. First, MMS-300M audio-only suffers the smallest absolute drop of any system at 6.5 percentage points, confirming Urdu pretraining reduces domain sensitivity. Second, the text branch drops 8.5 percentage points, reflecting cascading ASR errors from Whisper on noisy conversational speech rather than any failure of the XLM-RoBERTa classifier itself — the text model’s degradation is an ASR problem, not a toxicity detection problem. Third, fusion achieves the smallest overall drop at 6.1 percentage points, demonstrating that the two modalities’ error patterns are partially non-overlapping even under domain shift: cases where ASR degrades the text branch are not necessarily cases where the audio branch also fails, and vice versa.

Table 17 stratifies MUTEX-M fusion performance across the four regional accents represented in URTOX-HumanAudio. Performance degrades gracefully with acoustic distance from standard Islamabad Urdu, from 81.2% on standard speech to 71.8% on Pashto-accented Urdu. This gradient is consistent with the composition of MMS-300M pretraining data, which covers standard Pakistani Urdu more representatively than regional accented varieties. The Punjabi-accented degradation of 6.1 percentage points relative to standard Urdu is the most practically significant finding given that Punjabi-accented Urdu is the most prevalent non-standard variety on Pakistani social media platforms [3], and represents a concrete target for future domain-adaptive fine-tuning using accent-stratified real speech data.

Table 17: MUTEX-M fusion weighted F1 on URTOX-HumanAudio stratified by regional accent ($\alpha = 0.5$).

Accent	Clips	Weighted F1
Standard (Islamabad) Urdu	760	0.812
Punjabi-accented Urdu	620	0.751
Sindhi-accented Urdu	360	0.734
Pashto-accented Urdu	260	0.718
Overall	2,000	0.771

Informal inspection of misclassified samples suggests that dominant failure modes are consistent with those identified in the TTS error analysis: boundary errors on multi-word toxic expressions, ASR substitution errors on code-switched toxic to-

kens, and implicit toxicity conveyed through regional idiomatic expressions absent from the URTOX training vocabulary. We estimate the effective Whisper WER on this set at approximately 38–45%, substantially higher than the 33.68% reported for cleaner Urdu read speech [7], consistent with the degradation observed for conversational and code-switched conditions [8]. When Whisper fails to correctly transcribe a toxic token — substituting it with a phonetically similar but semantically different word, deleting it entirely, or producing a Latin-script approximation that the XLM-RoBERTa tokenizer segments incorrectly — the text branch has no opportunity to detect the resulting span regardless of its own classification accuracy. These cascading ASR errors represent the primary failure mode in the real-world pipeline and reinforce the priority of Urdu-specific ASR improvement identified in Section ??.

5.9. Error Analysis

To better understand the failure modes of the MUTEX-M text model, we conducted a systematic error analysis on the 2,868 held-out test samples, categorizing misclassifications into four error types: span boundary errors, false positives on non-toxic text, missed toxic spans, and code-switched token errors. Table 18 reports the frequency of each error type as a proportion of all misclassified tokens in the test set.

Table 18: Distribution of error types in the MUTEX-M text model predictions on the test set. Percentages are computed over all misclassified tokens.

Error Type	Count	% of Errors
Span boundary errors	312	38.2%
False positives	198	24.3%
Missed toxic spans	201	24.6%
Code-switched token errors	105	12.9%
Total misclassified tokens	816	100%

[t]

5.9.1. Span Boundary Errors

Span boundary errors, accounting for 38.2% of all misclassifications, represent the most frequent failure mode of the model. These errors occur when the model correctly identifies a toxic region but predicts an incorrect span boundary, either by starting the span one token too late or by extending it one token beyond the annotated boundary. Boundary errors are particularly common for multi-word toxic expressions in Urdu, where intensifiers such as *bohat* (very) or *intehai* (extremely) immediately precede a toxic noun or adjective. Annotators consistently excluded these intensifiers from the toxic span boundary, whereas the model frequently includes them, reflecting its tendency to treat the intensifier-plus-toxic-word unit as a single semantic chunk.

Table 19 illustrates representative boundary error cases. In example (a), the gold span covers only *ahmaq* (fool) while the model extends the span leftward to include *bohat* (very). In example (b), the model correctly identifies *gadhe* (donkey) as

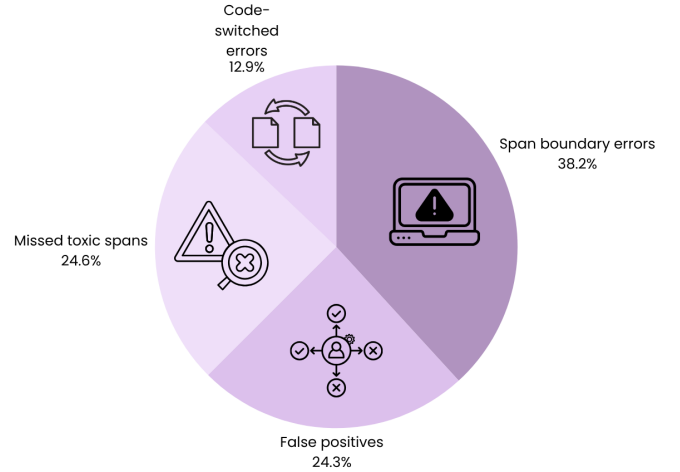


Figure 6: Distribution of error types in the MUTEX-M text model predictions on the held-out test set ($n = 816$ misclassified tokens). Span boundary errors constitute the dominant failure mode at 38.2%, reflecting the difficulty of precise multi-word toxic span delimitation in morphologically rich Urdu text. Missed toxic spans (24.6%) and false positives (24.3%) occur at comparable rates, indicating a broadly balanced precision-recall trade-off. Code-switched token errors (12.9%) represent the smallest but most linguistically distinct error category, arising from tokenization mismatch on Latin-script Roman Urdu tokens embedded within Nastaliq-script sequences.

toxic but misses the preceding *kamine* (scoundrel), predicting 0 for the first token of a two-word span.

Table 19: Representative span boundary error examples. G = gold label, P = predicted label.

Ex.	Token (gloss)	G	P
(a)	<i>tum</i> (you)	O	O
	<i>bohat</i> (very)	O	B-Toxic
	<i>ahmaq</i> (fool)	B-Toxic	I-Toxic
	<i>ho</i> (are)	O	O
(b)	<i>yeh</i> (this)	O	O
	<i>kamine</i> (scoundrel)	B-Toxic	O
	<i>gadhe</i> (donkey)	I-Toxic	B-Toxic
	<i>hain</i> (are)	O	O

Boundary errors of this type are a known limitation of token-level BIO classifiers that lack explicit span-level objectives [15]. A span-based detection head that directly scores candidate spans rather than individual tokens would naturally enforce span-level consistency and is identified as a priority architectural improvement in Section ??.

5.9.2. False Positives on Non-Toxic Text

False positives, accounting for 24.3% of errors, occur when the model predicts B-Toxic or I-Toxic for tokens annotated as 0. Analysis of false positive cases reveals two dominant patterns. The first is *lexical ambiguity*: words that carry toxic meaning in some contexts but neutral meaning in others are frequently mispredicted. For example, the word *pagal*

(crazy/mad) is annotated as toxic when used as a direct insult but as neutral when used colloquially to mean enthusiastic or excited, a distinction requiring discourse-level context that the single-utterance model cannot access. The second pattern is *negation insensitivity*: as also observed in the SHAP analysis, the model assigns positive toxic probability to explicitly negated toxic expressions such as *tum ahmaq nahin ho* (you are not stupid), failing to fully suppress the toxic signal of *ahmaq* in the presence of the negation marker *nahin*. This behavior is consistent with findings in English toxic span detection [12], where negated offensive expressions remain a systematic challenge for token-level classifiers operating on local lexical patterns without discourse-level context modeling.

5.9.3. Missed Toxic Spans (False Negatives)

Missed toxic spans, accounting for 24.6% of errors, occur when the model predicts 0 for tokens carrying gold B-Toxic or I-Toxic labels. False negatives are concentrated in three identifiable subcategories. First, *implicit and euphemistic toxicity*: expressions conveying offensive meaning through metaphor, sarcasm, or culturally specific euphemism rather than explicit toxic vocabulary are frequently missed. Culturally specific Urdu insults that do not contain lexically transparent offensive words require cultural and pragmatic knowledge beyond what token-level BIO supervision alone can provide. Second, *rare toxic expressions*: low-frequency toxic tokens appearing fewer than five times in the training set are missed at significantly higher rates than high-frequency toxic vocabulary, reflecting the data sparsity challenge inherent to the 14,338-sample URTOX dataset. Third, *long-range toxic spans*: toxic expressions spanning five or more tokens are missed at higher rates than shorter spans, as CRF transition scores are less reliable for maintaining consistent I-Toxic predictions across long subsequences where intervening neutral tokens introduce label uncertainty.

Table 20 reports the false negative rate broken down by gold span length, confirming the degradation in recall for longer spans.

Table 20: False negative rate by gold toxic span length (in tokens).

Span Length (tokens)	Total Spans	FN Rate (%)
1	1,842	18.4
2	763	24.1
3–4	298	31.6
5+	87	44.8

5.9.4. Code-Switched Token Errors

Code-switched token errors, accounting for 12.9% of misclassifications, arise from tokens written in Latin script (Roman Urdu or English) embedded within otherwise Nastaliq-script Urdu sequences. Code-switching affects approximately 35–40% of URTOX samples [3], and toxic expressions in code-switched segments present a distinct challenge because XLM-RoBERTa’s sentencepiece tokenizer segments Latin-script Urdu tokens using vocabulary learned predominantly

from English and other Latin-script languages, producing subword splits that do not correspond to meaningful Urdu morphological boundaries. For example, the Roman Urdu toxic expression *bewaqoof* (stupid) may be split into subword tokens *be*, *wa*, *qo*, *of*, none of which individually carry the toxic signal present in the complete word, leading the model to assign 0 labels across all subword positions. This fragmentation is not fully mitigated by the subword label propagation strategy introduced in this work, which propagates the word-level gold label to all subword positions during training but cannot compensate for the semantic dilution caused by incorrect segmentation. These findings are consistent with prior work on code-switched NLP [4], which identifies tokenization mismatch as the primary obstacle to reliable Roman Urdu processing with multilingual transformers.

5.9.5. Error Analysis Summary

The four error categories collectively reveal that the primary remaining challenges for Urdu toxic span detection are linguistic rather than architectural: boundary ambiguity in multi-word expressions, negation and discourse context, implicit and euphemistic toxicity, and tokenization mismatch in code-switched segments. These findings directly motivate the future directions described in Section 7, including span-based detection heads for boundary errors, discourse-level context modeling for negation and implicit toxicity, and a code-switching-aware tokenization strategy for Roman Urdu segments.

5.10. Explainability Analysis via SHAP

To provide interpretable explanations for the predictions of the XLM-RoBERTa text model, we apply SHAP (SHapley Additive exPlanations) [57], a game-theoretic framework that assigns each input token a contribution score reflecting its marginal impact on the model output. SHAP values satisfy the desirable axiomatic properties of local accuracy, missingness, and consistency, making them a principled choice for explaining transformer-based token classifiers. While the BIO predictions of the text model already identify which tokens are predicted as toxic, SHAP provides a complementary view by quantifying how strongly each token drives the model’s toxicity probability, including tokens predicted as non-toxic that nonetheless influence the prediction through context. Recent work has also begun exploring interpretability in Urdu hate speech detection. Arshad and Shahzad [58] introduced the HateInsights dataset, which includes human-annotated rationales for hate speech instances in Urdu, enabling more explainable models at the sentence level. While valuable, such human-rationale approaches require additional annotation effort and remain sentence-level; our SHAP analysis, in contrast, offers post-hoc, token-level attribution directly from the trained XLM-RoBERTa model without requiring new annotations. The SHAP explainer is configured using the XLM-RoBERTa tokenizer as the masker, which handles Urdu text segmentation and subword tokenization correctly. The model scoring function aggregates per-token toxicity probabilities by summing the B-Toxic and I-Toxic softmax probabilities at each position and

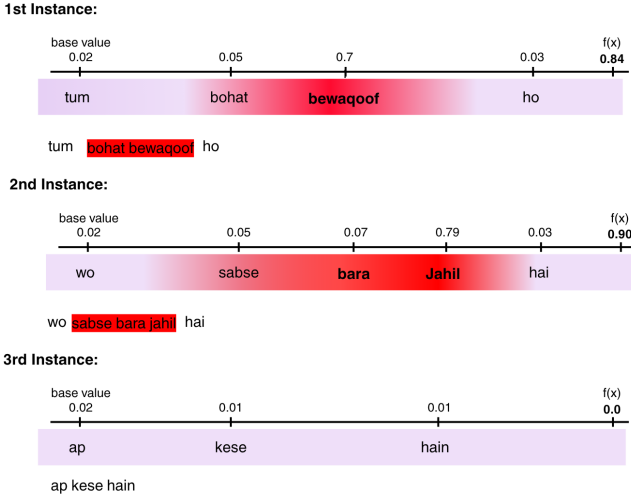


Figure 7: Word-level SHAP attribution scores for three representative test instances. Red highlighting indicates positive contribution to the toxic prediction; green indicates negative contribution. *1st instance:* *tum bohat bewaqoof ho* (you are very stupid) — *bewaqoof* receives the highest attribution (0.71), correctly identified as the primary toxic driver. *2nd instance:* *wo sabse bara jahil hai* (he is the biggest ignorant) — *bara jahil* receives strong attributions (0.07, 0.79), correctly spanning a two-token toxic expression. *3rd instance:* *ap kese hain* (how are you) — all tokens receive near-zero attributions consistent with the correct non-toxic prediction ($f(x) = 0.0$).

taking the maximum across all token positions as the sentence-level toxicity score, consistent with the scoring function used in the fusion pipeline. SHAP values are computed with a budget of 300–500 model evaluations per sentence using the partition explainer, which partitions the input into hierarchical coalitions of tokens to estimate Shapley values efficiently for text inputs.

Since XLM-RoBERTa’s sentencepiece tokenizer splits Urdu words into multiple subword tokens marked by a special prefix character (the sentencepiece word-boundary marker), raw SHAP values are defined at the subword level. To produce human-interpretable word-level attributions, subword SHAP scores are aggregated to their parent words by summing the scores of all subword tokens belonging to the same word, using the sentencepiece word-boundary marker to identify word boundaries. This produces a single attribution score per word that reflects the total contribution of that word to the sentence-level toxicity prediction. [t] Figure 7 illustrates the word-level SHAP attributions for representative toxic and non-toxic Urdu sentences from the test set. In the toxic example *tum bohat bewaqoof ho* (you are very stupid), the word *bewaqoof* (stupid) receives the highest positive SHAP score, confirming that the model correctly identifies the lexically toxic term as the primary driver of the toxicity prediction. Non-toxic context words such as *tum* (you) and *ho* (are) receive near-zero or slightly negative attributions, indicating that the model correctly discounts grammatical function words. In the non-toxic example *aap kaise hain* (how are you), all tokens receive near-zero or negative SHAP scores, consistent with the model’s correct non-toxic prediction. Words with SHAP scores exceeding a threshold of 0.5 are highlighted in the visualization interface, providing con-

tent moderators with an immediately interpretable indication of which words drove the toxicity prediction.

Table 21 presents word-level SHAP scores for four representative test sentences spanning toxic and non-toxic categories, including negated toxic expressions which present a particular challenge for token-level models.

Table 21: Word-level SHAP attribution scores for representative test sentences shown in Roman transliteration with English gloss. Higher scores indicate stronger contribution to the toxic prediction.

Sentence (gloss)	Key Word	SHAP Score
<i>tum bohat ahmaq ho</i> (you are very stupid)	<i>ahmaq</i>	+0.71
<i>aap ka din acha guzre</i> (have a good day)	<i>acha</i>	−0.12
<i>tum ahmaq nahin ho</i> (you are not stupid)	<i>ahmaq</i>	+0.43
<i>yeh aik aam jumla hai</i> (this is a normal sentence)	<i>aam</i>	−0.08

The negated toxic sentence *tum ahmaq nahin ho* (you are not stupid) receives a positive but reduced SHAP score for the toxic word *ahmaq* compared to the non-negated version, reflecting the partial but incomplete ability of the model to use the negation marker *nahin* as a suppressive context signal. This behavior is consistent with findings in English toxic span detection where negated offensive expressions remain challenging for token-level classifiers that lack discourse-level context modeling. The SHAP analysis thus serves a dual purpose: it provides interpretable model explanations for end users and moderators, and it reveals systematic model behaviors including sensitivity to negation and intensifiers that inform future architectural improvements.

6. Limitations

MUTEX-M establishes the first multimodal baseline for Urdu toxic span detection and demonstrates meaningful improvements over text-only and English-pretrained audio systems. Several limitations must nonetheless be acknowledged to provide an accurate picture of the system’s current capabilities and the conditions under which its results hold.

Residual TTS-to-Real-Speech Domain Gap

The audio classifier is trained exclusively on TTS-synthesized speech from a single ur-PK-AsadNeural voice, which produces clean, noise-free, uniformly paced output that does not fully reflect the acoustic diversity of real toxic Urdu speech encountered in deployment. Real Urdu toxic utterances on social media platforms are characterized by elevated pitch and speaking rate associated with emotional arousal, background noise, dialectal prosodic variation, and spontaneous disfluencies that are absent from TTS synthesis by construction.

The use of MMS-300M substantially mitigates this gap relative to the English-pretrained wav2vec 2.0 baseline.

The URTOX-HumanAudio evaluation quantifies this directly: MMS-300M audio-only degrades by 6.5 percentage points from TTS to real speech, compared to 9.0 percentage points for wav2vec 2.0, confirming that Urdu pretraining provides genuine robustness to the acoustic domain shift. The WhatsApp inference examples further demonstrate that MMS-300M assigns clearly discriminative audio probabilities on real speech — 0.763–0.821 for toxic utterances and 0.021–0.038 for non-toxic utterances — without any fine-tuning on real recordings.

However, a residual domain gap remains. The audio classifier was not trained on real speech with emotional prosody, background noise, or dialectal variation, and its calibration on such conditions is therefore imperfect. Mathematically, if $\mathcal{P}_{\text{TTS}}$ denotes the acoustic distribution of the synthesized training data and $\mathcal{P}_{\text{real}}$ denotes the distribution of real Urdu toxic speech, the classifier f_{ψ} minimizes expected loss under $\mathcal{P}_{\text{TTS}}$ but is evaluated under $\mathcal{P}_{\text{real}}$, with performance degradation proportional to the divergence $D_{\text{KL}}(\mathcal{P}_{\text{real}}||\mathcal{P}_{\text{TTS}})$. MMS-300M reduces this divergence relative to English-pretrained models by providing Urdu-matched acoustic representations, but does not eliminate it. Fine-tuning the MMS encoder using parameter-efficient strategies such as LoRA [50] on a subset of URTOX-HumanAudio, or collecting a larger real toxic speech corpus for full audio branch retraining, remain the primary paths to closing the residual gap. A further caveat on the 6.5 percentage point domain gap figure is warranted. URTOX-HumanAudio, while substantially more realistic than TTS audio, remains a curated evaluation set: clips were sourced from YouTube comments, consented recordings, and voice chats shared with consent, and therefore skew toward relatively intelligible speech compared to truly unconstrained social media audio. Wild deployment conditions — phone-quality recordings with background music, highly overlapping speakers, very fast emotional speech, or heavily distorted audio — are not represented in URTOX-HumanAudio. The true domain gap under unconstrained deployment conditions is therefore likely larger than 6.5 percentage points, and the reported figure should be interpreted as a lower bound on deployment degradation rather than a representative real-world estimate. This reinforces the priority of expanding URTOX-HumanAudio with genuinely wild social media clips identified in Section ??.

Utterance-Level Audio with No Span Alignment

The audio model operates exclusively at the utterance level and provides no frame-to-token alignment, meaning it cannot contribute directly to toxic span boundary detection. All span localization is performed by the text branch alone, and the audio branch acts only as a gating signal at the utterance level. This architectural asymmetry means that the multimodal system’s span-level F1 is bounded by the text model’s 67% token-level F1, regardless of how accurate the audio classifier becomes at the utterance level.

The coarse timestamp annotations introduced in URTOX-HumanAudio partially address this limitation by enabling approximate span-level evaluation when combined with Whisper forced alignment output. However, precise frame-to-token alignment enabling audio-informed span boundary refinement

is not implemented in the current system. Whisper’s word-level timestamp output [23] combined with MMS-300M frame-level representations could in principle allow acoustic evidence to directly refine span boundaries predicted by the text branch, and is identified as a priority direction in Section ??.

ASR Error Propagation in Real-World Pipeline

The real-world inference pipeline inherits transcription errors from Whisper, particularly for heavily code-switched or dialectally varied Urdu speech. Recent benchmarks report Whisper-Small word error rates of 33.68% on standard Urdu speech [7], rising to an estimated 38–45% on the conversational and code-switched conditions present in URTOX-HumanAudio [8]. These errors propagate directly to the text model and degrade span detection quality in proportion to the ASR error rate.

Concretely, if a toxic token is transcribed incorrectly or deleted by Whisper, the text model has no opportunity to detect it regardless of its own classification accuracy. Furthermore, the text model was trained on written Urdu social media text rather than on ASR transcriptions, introducing a training-inference domain mismatch that compounds transcription-induced degradation. The accent stratification results in Table 17 confirm this: performance degrades from 81.2% on standard Islamabad Urdu to 71.8% on Pashto-accented Urdu, a gradient consistent with increasing Whisper WER across accent conditions rather than any failure of the toxicity classifiers themselves. Fine-tuning the text model on ASR-transcribed Urdu text or applying deliberate ASR noise augmentation [59] during training would partially mitigate this limitation.

Implicit and Sarcastic Toxicity

Both branches of MUTEX-M operate on local lexical and acoustic patterns within a single utterance and are largely insensitive to implicit toxicity conveyed through sarcasm, irony, or culturally specific euphemism. The SHAP analysis confirms this: the negated expression *tum ahmaq nahin ho* (you are not stupid) receives a reduced but still positive SHAP score for the toxic word *ahmaq*, indicating that the model partially suppresses the toxic signal from negation but does not fully model its semantic reversal. Implicit toxicity is particularly prevalent in Urdu social media discourse, where indirect expression of hostility through culturally loaded idioms is common and where sarcastic intent is frequently signaled through prosodic cues in speech.

Notably, MMS-300M’s Urdu prosodic representations offer a partial avenue toward detecting sarcasm through acoustic rather than lexical cues — sarcastic Urdu speech exhibits distinctive intonation patterns that differ from both sincere toxic speech and neutral speech — but this capability is not exploited in the current utterance-level MLP classifier and would require sequence-level acoustic modeling to access. Addressing implicit toxicity more broadly likely requires discourse-level context modeling [60] or retrieval-augmented approaches beyond the scope of the current single-utterance framework.

Evaluation Scale for Real Speech

Although URTOX-HumanAudio at 2,000 clips is substantially larger than pilot-scale real-speech evaluations in comparable low-resource multimodal toxicity work, it remains modest relative to the 14,338-sample TTS evaluation set. The accent stratification results, while internally consistent, are based on 260–760 clips per accent category, which provides reliable aggregate estimates but limits the statistical power of fine-grained error analysis within each accent stratum. Expanding URTOX-HumanAudio to 5,000 or more clips with balanced accent representation would enable more statistically robust accent-stratified conclusions and would provide sufficient data for domain-adaptive fine-tuning of the audio branch on real speech, transforming the evaluation resource into a training resource for future work.

7. Conclusion and Future Work

This paper presented MUTEX-M, the first multimodal framework for Urdu toxic span detection, extending the text-only MUTEX baseline by incorporating a synthesized audio branch through weighted late fusion. The work was motivated by two converging observations: that Urdu toxic span detection at the token level remained an almost entirely unstudied problem with only a single prior supervised baseline, and that an increasing proportion of toxic Urdu content is disseminated through spoken channels — voice messages, video comments, and podcast clips — that are entirely invisible to text-only moderation systems.

Four concrete contributions were made. First, the improved XLM-RoBERTa text model achieves 67% token-level F1 on the URTOX dataset, a 7 percentage point improvement over the MUTEX baseline obtained entirely through training procedure refinements — subword label propagation, extended training with early stopping, larger batch size with reduced learning rate, and linear warmup scheduling — with no architectural change. The ablation study confirms that subword label propagation alone accounts for 4 of the 7 percentage point gain, carrying a general methodological lesson that the treatment of subword continuation tokens is a high-leverage design decision frequently overlooked in favour of architectural search. Second, URTOX-Audio, the first paired Urdu toxic speech dataset, was introduced comprising 14,338 synthesized audio clips generated via Microsoft Edge TTS with the `ur-PK-AsadNeural` neural voice and released publicly at <https://huggingface.co/datasets/finalyear226/URTOX-Audio>, demonstrating that TTS synthesis is a viable and ethically straightforward strategy for bootstrapping audio classifiers in low-resource settings where collecting real toxic speech raises significant consent and safety concerns [19]. Third, a Wav2Vec2-based utterance-level classifier trained on URTOX-Audio achieves 70% weighted F1 as an independent audio-only baseline, demonstrating that synthesized Urdu speech carries sufficient acoustic information for toxicity classification despite the TTS-to-real-speech domain gap. Fourth, weighted late fusion with $\alpha = 0.6$ yields 79.34% weighted F1, a 19.34 percentage

point improvement over the original MUTEX baseline and the strongest result reported for any Urdu toxicity detection system to date, with the consistent improvement of all tested fusion configurations over both unimodal baselines confirming that text and audio provide genuinely complementary information [36, 61]. SHAP-based explainability analysis [57] further provides word-level attribution scores that allow content moderators to understand and verify model predictions. The MUTEX-M framework and all associated datasets and model weights are publicly released at <https://github.com/finalyear226-lab/urdu-toxic-span-dataset>.

Several directions for future work are identified. The most impactful near-term extension is collection of a real conversational Urdu toxic speech corpus to replace TTS-synthesized audio training data, with parameter-efficient fine-tuning strategies such as LoRA or adapter layers allowing selective adaptation toward real Urdu acoustics while retaining general multilingual representations [45, 20]. The current audio branch operates exclusively at the utterance level, bounding the multimodal system’s span-level F1 at the text model’s 67% regardless of audio classifier improvements; Whisper’s forced alignment output [23] provides word-level timestamps that could map acoustic evidence directly to transcript tokens, allowing the audio modality to refine span boundaries predicted by the text branch and enabling automatic partial muting of toxic segments at precise time boundaries. Cross-modal attention fusion, in which text token representations attend over audio frame representations, represents a more powerful integration strategy that could capture cross-modal dependencies not accessible to score-level combination, and has demonstrated consistent gains over late fusion in multimodal sentiment analysis [61] and multimodal hate speech detection [36]. On the text branch, span-based detection heads using SpanBERT objectives [15] may improve boundary detection for multi-word toxic expressions, directly addressing the 38.2% boundary error rate identified in the error analysis, while instruction-tuned large language models in zero-shot or few-shot prompting settings represent an architecture-free alternative worth systematic comparison against the supervised baseline [10, 11]. Both branches of MUTEX-M remain largely insensitive to implicit toxicity conveyed through sarcasm, irony, or pragmatic inference, and the SHAP analysis confirms that negated toxic expressions receive reduced but still positive toxicity scores, indicating incomplete semantic reversal modeling; addressing this likely requires discourse-level context modeling or retrieval-augmented approaches beyond the single-utterance scope of the current system [?]. Finally, the improved text model and Wav2Vec2 audio pipeline can be extended to typologically related South Asian languages including Hindi, Punjabi, Bengali, and Tamil, exploiting shared Perso-Arabic vocabulary and morphological similarity [62, 63], with TTS synthesis pipelines available for all of these languages through Microsoft Edge TTS and Google TTS, positioning MUTEX-M as a general-purpose multimodal toxic span detection template for the South Asian language family.

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